Combinatorial algebra

The problems we will discuss today are accused by some of being "combinatorics". Most of them are clearly more algebra than anything else, but they often involve rather arbitrary sets or sequences of numbers, and generally do not require heavy computation. They may ask to show that something "exists" or "is possible". Also, they tend to use a way of thinking that we normally associate with combinatorics: for example, some extremal element, or Pigeonhole principle. However, maybe a few problems are here just because they didn't fit anywhere else.

1 Prove that any 101 numbers $0 = x_0 \leqslant x_1 \leqslant \cdots \leqslant x_{100} = 1$ can be divided into two groups such that their arithmetic means differ by at least $\frac{101}{200}$.

2 Prove that, among any 12 distinct numbers from the interval $(1, 12)$, there are three distinct numbers that are the side lengths of an acute-angled triangle.

3 Given any seven positive real numbers, prove that among them there are x and y such that $x > y$ and $1 + xy > (2 + \sqrt{3})(x - y)$.

4 Prove that among any nine distinct real numbers there exist four distinct numbers a, b, c, d such that $(a^2 + b^2)(c^2 + d^2) \leqslant 1.1 \cdot (ac + bd)^2$.

5 Some 100 numbers are arranged in a circle. Each number is greater than the sum of the two (clockwise) subsequent numbers. At most how many of these numbers can be positive?

6 Let $n > 1$ be an integer. Real numbers $x_1, x_2, \ldots, x_n$ add up to zero. At least how many pairs of indices $i < j$ are that for which $x_i + x_j \geqslant 0$?

7 Prove that in any set of 2000 distinct real numbers there exist two distinct pairs (a, b) and (c, d) (not necessarily disjoint) such that $\left|\frac{a-b}{c-d} - 1\right| < \frac{1}{100000}$.

8 Consider a partition of set $\{1, 2, \ldots, 100\}$ into pairwise disjoint nonempty subsets $S_1, S_2, \ldots, S_n$ for some n, i.e. Denote by a_i the average of elements of S_i. Determine the smallest possible value of the average of the a_i.

9 The sum of several real numbers from $[0, 1]$ does not exceed S. Find the maximum value of S such that it is always possible to partition these numbers into two groups with sums not greater than 9.

10 By $\{x\}$ we denote the fractional part of x. If $x_1, x_2, \ldots, x_n$ are any real numbers, prove that there exists a real number a such that $\{a + x_i\} \leqslant \frac{n-1}{n}$ for each i.

11 Let $x_1, x_2, \ldots, x_n$ be real numbers with the sum 0. Also define $x_{n+i} := x_i$. Prove that there exists k with $1 \leqslant k \leqslant n$ such that each of the sums x_k, x_k+x_{k+1}, $x_k+x_{k+1}+x_{k+2}$, $\ldots$, $x_k+x_{k+1}+\cdots+x_{k+n-1}$ is nonnegative. *(The gas station problem)*

12 Let $a_1, a_2, \ldots, a_n$ be a sequence of real numbers. We say a_i is a *bomb* if there is $j \geqslant i$ such that the arithmetic mean of $a_i, \ldots, a_j$ is at least 1. Prove that the arithmetic mean of all bombs is at least 1.

13 Let S be a set of positive numbers with the sum 1. We will call an element $x \in S$ a *knife* if the set $S \setminus \{x\}$ can be divided into two subsets with sums of elements at most $\frac{1}{2}$. Prove that the sum of all knifes is at least $\frac{1}{2}$.

14 The sum of some $2n - 1$ nonnegative real numbers equals 1. Consider all sums of n of these numbers. At least how many of these sums can be not less than $\frac{1}{2}$?

15 Let $A = (a_{ij})$ be an $n \times n$ matrix, where a_{ij} are nonnegative integers. Whenever $a_{ij} = 0$, the sum of the elements in the i-th row and the j-th column is at least n. Prove that the sum of all elements in the matrix is at least $\frac{n^2}{2}$.

16 Positive real numbers are written in a $2 \times n$ array so that the sum in each of the n column equals 1. Prove that we can always select a number in each column so that the sum of the selected numbers in each row is at most $\frac{n+1}{4}$.

17 Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of real numbers. Suppose that each term a_i belongs to a strictly monotonic segment of length at least $i+1$. Prove that the entire sequence is strictly monotonic starting from some point.

18 For which n is it possible to wrote n distinct numbers around a circle so that every number is either greater than the next 100 or less than the next 100, but so that this property is lost whenever one number is removed?

19 Given real numbers $a_1, a_2, \ldots, a_n$, define $d_i = \max\{a_1, \ldots, a_i\} - \min\{a_{i+1}, \ldots, a_n\}$ for $1 \leqslant i \leqslant n-1$. Let $d = \max d_i$. Find the smallest h for which there exist real numbers $x_1 \leqslant x_2 \leqslant \ldots x_n$ such that $|x_i - a_i| \leqslant h$ for all i.

20 Let $x_1, \ldots, x_n$ be real numbers. Order them as $y_1, \ldots, y_n$ inductively: given $y_1, \ldots,\ y_{i-1}$, choose y_i so that $|y_1 + \cdots + y_i|$ is minimal. Find the largest c such that we always have $\max_k |x_1 + \cdots + x_k| \geqslant c \cdot \max_k |y_1 + \cdots + y_k|$.

21 Prove that for any n positive numbers $a_1, a_2, \ldots, a_n$ one can find positive numbers $b_1, b_2, \ldots, b_n$ such that $b_i \geqslant a_i$ and, whenever $b_i \leqslant b_j$, b_j/b_i is an integer, and moreover, $b_1 \cdots b_n \leqslant 2^{\frac{n-1}{2}} a_1 \cdots a_n$.

22 We say that a sequence of nonnegative real numbers $a_1, a_2, \ldots, a_k$ is *embeddable* in the interval $[b, c]$ if there exist numbers $x_0, x_1, \ldots, x_k \in [b, c]$ such that $|x_i - x_{i-1}| = a_i$ for $i = 1, 2, \ldots, k$. The sequence is *normalized* if its terms do not exceed 1. Given a positive integer n, prove that:

(a) every normalized sequence of length $2n+1$ is embeddable in the interval $[0, 2-\frac{1}{2^n}]$;

(b) for $n > 1$, not every normalized sequence of length $4n-1$ is embeddable in the interval $[0, 2-\frac{1}{2^n}]$.

23 By an *extension* of an interval $[a, a+d]$ we mean the interval $[a-\frac{d}{2}, a+\frac{3d}{2}]$. Prove that there exists a positive integer m with the following property: Every finite set A of real numbers has a subset B with at most m elements such that, whenever B is covered with 100 intervals, A is covered with their extensions.

24 Let $n \geqslant 2$ be an integer. Find the smallest real number λ with the following property: for any real numbers $x_1, x_2, \ldots, x_n \in [0, 1]$ there exist numbers $z_i \in \{x_i, x_i - 1\}$ such that $|\sum_{i=a}^{b} z_i| \leqslant \lambda$ for all $1 \leqslant a \leqslant b \leqslant n$.

25 If $1 \leqslant a_1 \leqslant a_2 \leqslant \ldots \leqslant a_n \leqslant n$ are integers, prove that $\sum_{i=1}^{n} (a_{a_i} - a_i) \leqslant \frac{(n-1)^2}{4}$.

26 Baron Munchausen ordered a thin black-and-white wooden strip of length 1 whose white and black parts have equal total lengths. Due to a bug, the carpenter produced a strip with the same coloring pattern, but shrunk to half the required size; thus, to make up for his mistake, he produced an identical strip and handed both to the baron. The baron then claimed that he managed to glue the two strips and recolor several segments of total length at most 0.01 so as to obtain the exact coloring pattern he wanted. Is this possible?